BUS-F420: Equity and Fixed Income Investments

Lecture 12: Valuing Debt Instruments

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Recap from last time

- Last time, talked entirely about loans
- **Three types:**
  1. Fully amortizing $\rightarrow$ loan paid off over life of the loan
  2. Balloon loans $\rightarrow$ balloon payment at the end
  3. Interest-only loans $\rightarrow$ "lock-in period" with interest-only payments

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- This is, essentially, a fixed-coupon bond!
  - Also known as *bullet bonds* or *straight bonds*

- **Key features**
  - Repayment only occurs at maturity
  - Coupon (payment) rate is fixed

- Because there's no amortization, calculating interest payments is easy

$$P = \begin{cases} i \times L, & \text{before maturity} \\ i \times L + L, & \text{at maturity} \end{cases}$$

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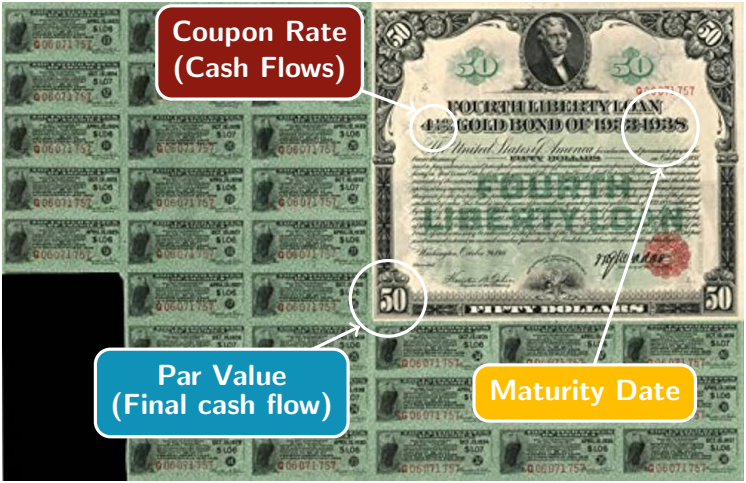

What is the value of an asset?

- All (cash-generating) assets can be valued by the same equation:

$$\underset{\text{(Price)}}{\text{Value}} = V = \frac{CF_1}{1+y} + \frac{CF_2}{(1+y)^2} + \cdots + \frac{CF_T}{(1+y)^T}$$

$$= \sum_{t=1}^{T} \frac{CF_t}{(1+y)^t} \rightarrow \text{(Present value of future cash flows)}$$

- What are the individual cash flows (CF)?
  - Equity: future dividends
  - Bonds: future coupon payments

Valuing a "plain vanilla" debt instrument

- Let's begin by connecting a bond's features to notation:

M	—	par value
r	—	coupon rate
C	—	coupon payment ($= r \times M$)
T	—	periods until maturity
y	—	required yield

- Corporate bonds have a <u>semi-annual</u> coupon schedule
  - $T = 2 \times$ number of years
  - $r =$ annual rate/2
  - (Not always the case — ABS often monthly)

Valuing a "plain vanilla" debt instrument

- Using our valuation equation, the cash flows of a bond satisfy

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$$= C\left[\frac{1-(1+y)^{-T}}{y}\right] + \frac{M}{(1+y)^{T}}$$

- The coupons essentially form an *annuity*
  - **BUT:** it does not form the *entirety* of the value
  - **This is important to understand, and we'll emphasize this point repeatedly.**

A Numerical Example

- Let's work through an example to validate the annuity formula
- Consider a 2-year corporate bond with a 5% annual coupon rate:

M	—	par value	\$1,000
r	—	period coupon rate	2.5% (= 5% / 2)
C	—	period coupon payment	\$25 (= 2.5% × \$1,000)
T	—	periods until maturity	4
y	—	required yield	7%

Numerical Example: Coupons

- We can first use our valuation equation to calculate the PV directly:

$$\textbf{PV(Coupons)} = \frac{C}{(1+y)} + \frac{C}{(1+y)^2} + \frac{C}{(1+y)^3} + \frac{C}{(1+y)^4}$$

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Numerical example: Total

- Let's now combine the annuity value with the par value term:

$$\text{Price} = \mathbf{\$84.68} + \frac{\mathbf{\$1000}}{\mathbf{1.07^4}} = \$847.58$$

- What do you notice about the *price* relative to the *par value*?

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- Q: Any idea why?
  - A: coupon rate $<$ required yield

A second example with coupon > required yield

- Consider a **3-year** corporate bond with a **9%** annual coupon rate:

M	—	par value	\$1,000
r	—	semi-annual coupon rate	
C	—	semi-annual coupon payment	
T	—	periods until maturity	6
y	—	required yield	3%
		Annuity value:	
		PV of par value:	
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		Bond price:	\$1081.26

So what affects the price of a bond?

1. The underlying bond's *risk* is changing
   - Required rate is conditional on bond risk
   - We'll talk a lot about credit risk and pricing in the last part of the course

2. **Market rates (y) are changing**

3. **The bond is moving towards maturity ($T \downarrow$)**

What is the required yield?

- **Conceptually:** It is the current market yield of newly issued bonds
- Q: Suppose a bond has a coupon rate of 5%, while market yields are 7%. Which would you prefer?

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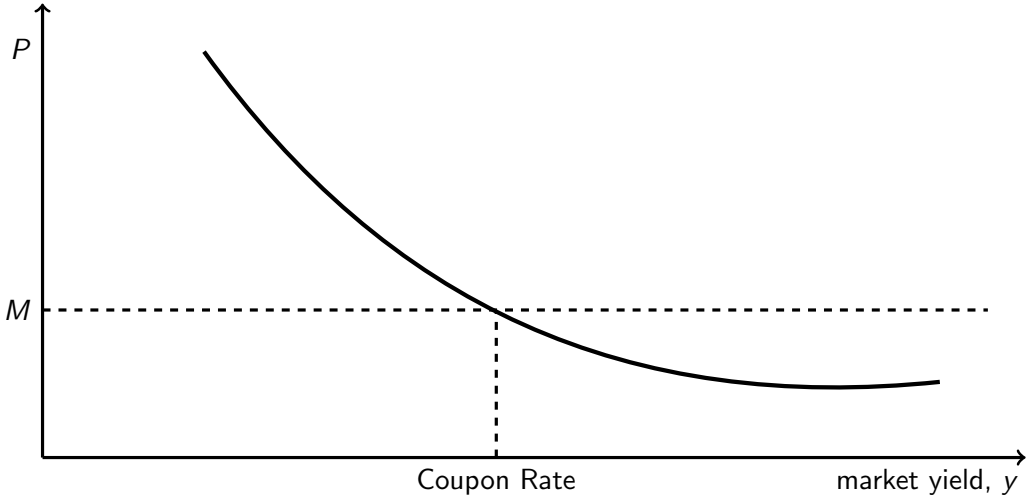
- Q: Suppose a bond has a coupon rate of 5%, while market yields are 7%. Which would you prefer?
  - A: Obviously the 7% bond.
  - Demand for 5% coupon bonds $\downarrow \rightarrow$ Price $\downarrow$!

- **Key idea:** Since price = par value at m'ty, gap resolves through **price appreciation**!

The Price-Yield Relationship for a Coupon Bond



Zero Coupon Bonds

- Some bonds do not pay explicit coupon payments — **zero coupon bonds**
- **Instead:** incorporate interest through *price appreciation*

$$\text{Price} = \frac{M}{(1+r)^T} \quad \leftarrow \quad \text{What happens as } T \text{ gets smaller?}$$

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- Many examples of zero-coupon bonds
  - Treasury strips
  - Municipal zeros

Compare a zero coupon bond to a coupon bond

	Bond 1	Bond 2
Par Value	\$1000	\$1000
Semi-annual Coupon Rate	5%	0%
Periods to maturity	4	4
Required yield	3%	3%
Price	\$1,074.34	\$888.49

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Approximate Return of Bond 1: $\underbrace{5\% \times 4}_{\text{coupons}} - \underbrace{7.4\%}_{\Delta\text{price}} = 12.6\%$

Approximate Return of Bond 2: $\dfrac{\$1000 - \$888.49}{\$888.49} = 12.6\%$

Floating-Rate Debt Valuation (Floaters)

- Everything so far has been about **fixed rate** debt
- **Floating rate** debt instead pegs coupon rates to a *reference rate*
  - Ex 1: 3 month T-bill
  - Ex 2: London interbank lending rate, SOFR (used to be LIBOR, which was phased out)
  - Often come with **maximum** or **minimum** coupon rates
- Floaters also typically pay a **spread** to attract investors
  - Typically fixed in the contract

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- Q: S'pose bond markets demand a lower spread. How might this manifest in floaters?
  - A: prices fall to permit price appreciation!
  - <u>Same idea as before:</u> prices can move to offset *fixed components* of contracts

Embeddable Bonds

- Think about an adjustable rate mortgage (ARM). Structured as:
  - Initial rate (e.g. 3% for 24 months)
  - Market rate (floating for remaining maturity of loan)

- Imagine if the owner of the mortgage (e.g. a bank) could *choose* to make rate floating
  - Q: What type of financial instrument is this akin to?

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- At the 24 month mark, when would the bank choose to exercise the option?
  - A: if market rate $>$ fixed rate

Types of Bonds with Embedded Options

- **Callable and putable bonds**
  - Gives a price in which issuer to retire/payoff the bond
  - Makes cash flows **uncertain** — valuation more difficult
- **Convertible bonds**
  - Gives bondholder right to convert to equity
  - Useful if issuer is growing!
  - However, these are often callable — even more nuance to pricing
- **Exchangeable bonds**
  - Similar to convertible bonds, but with another firm's equity

How do we value bonds with embedded options?

- Plain vanilla bonds are (somewhat) easy to value
  - Q: What makes our analysis so far possible?
  - A: No cash flow risk!
  - (Floaters have cash flow risk, but mostly through their peg)
- Let's pivot to bonds with **embedded options**
  - Recall: callable bonds, putable bonds
- Q: What features of these bonds make their cash flows uncertain?
  - A: The probability of exercising the option!
  - (Much different than a floater since this depends on bondholder behavior)

Logic of a Callable Bond's Price-Yield Relationship

- Let's think through the price-yield relationship of a **callable bond**
- **Remember:** A callable bond gives...
  - ...the *issuer* of the bond
  - ...the right to retire the bond
  - ...when the price is above some strike price
- Q1: If y is high, how should P compare to a (comparable!) option-free bond price?
- Q2: If y is low, how should P compare to a (comparable!) option-free bond price?

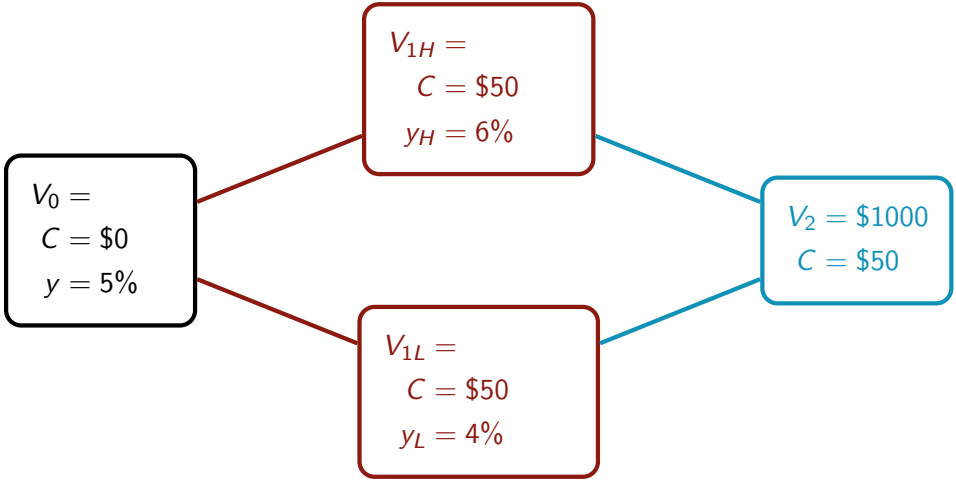
How can we value a callable bond?

- General framework very difficult for our purposes
  - (Need to model interest rate process, option values, etc.)

- **However:** we can use a simple binomial model to price the bonds!
  - Two states: high interest rate vs. low interest rate
  - Assumption 1: call price = par value
  - Assumption 2: bond is called if price > call price

- Assume two periods remaining in life of bond

How can we value a straight bond?



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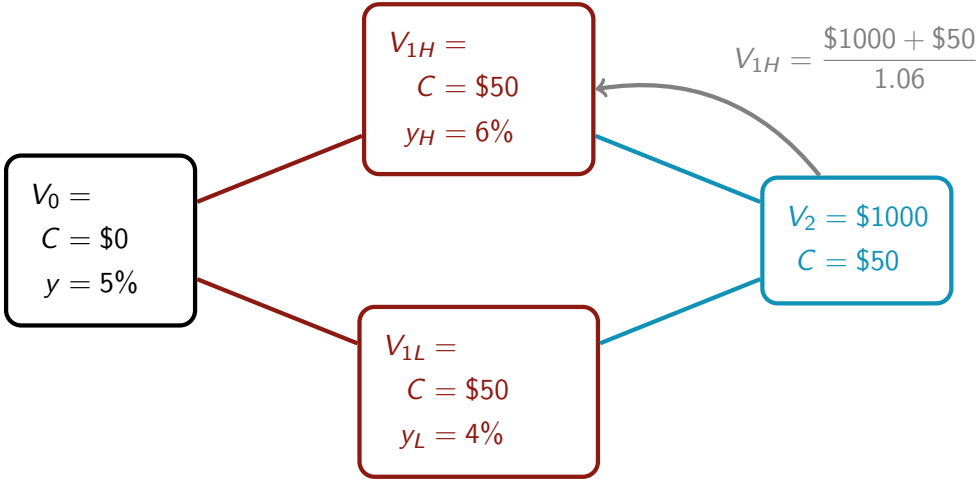

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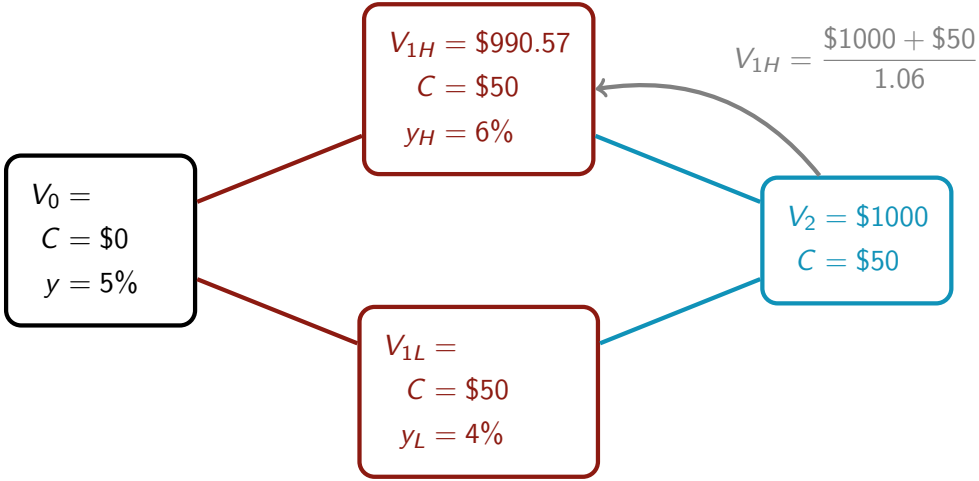

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$V_0 =$
$C = \$0$
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$V_{1H} = \$990.57$
$C = \$50$
$y_H = 6\%$

$V_{1L} =$
$C = \$50$
$y_L = 4\%$

$V_2 = \$1000$
$C = \$50$

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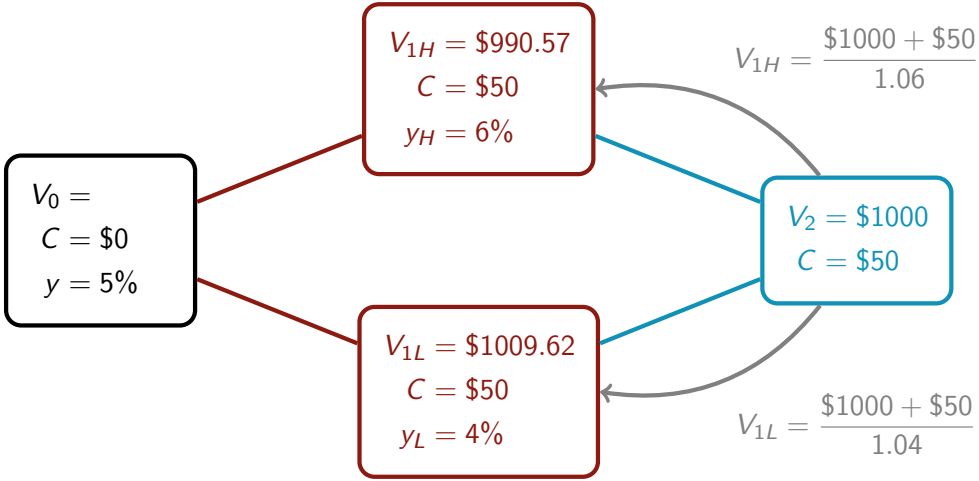

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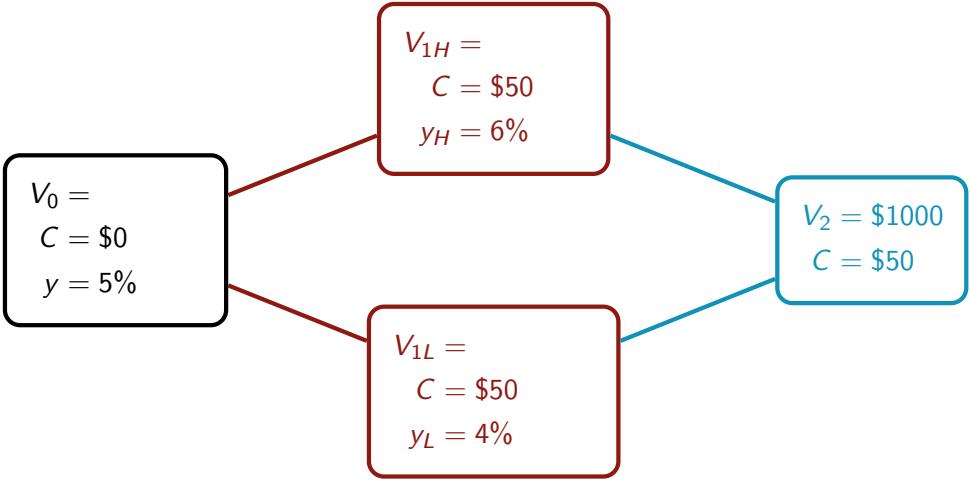

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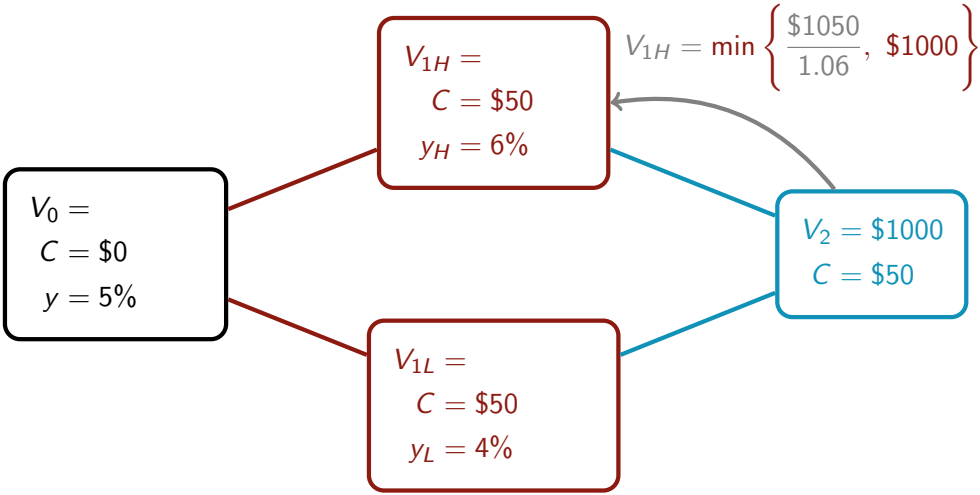

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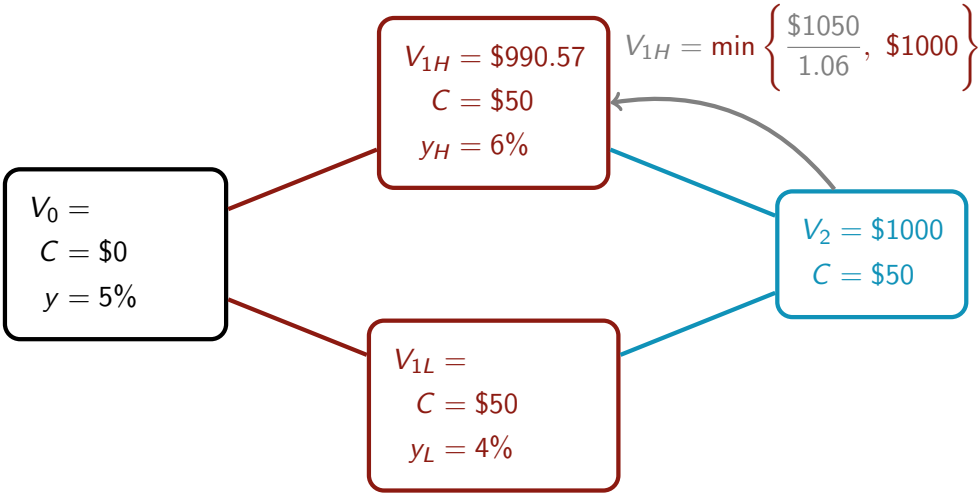

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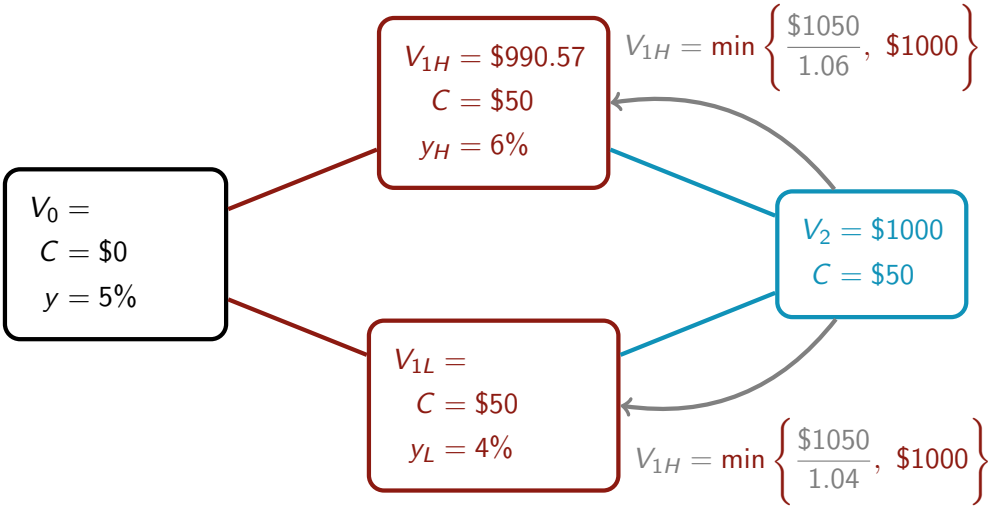

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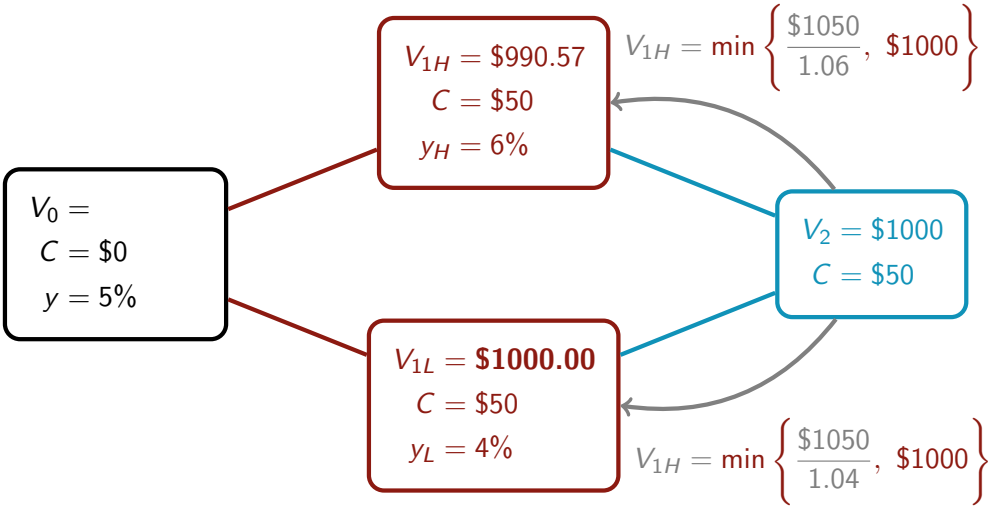

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$$V_2 = \$1000$$
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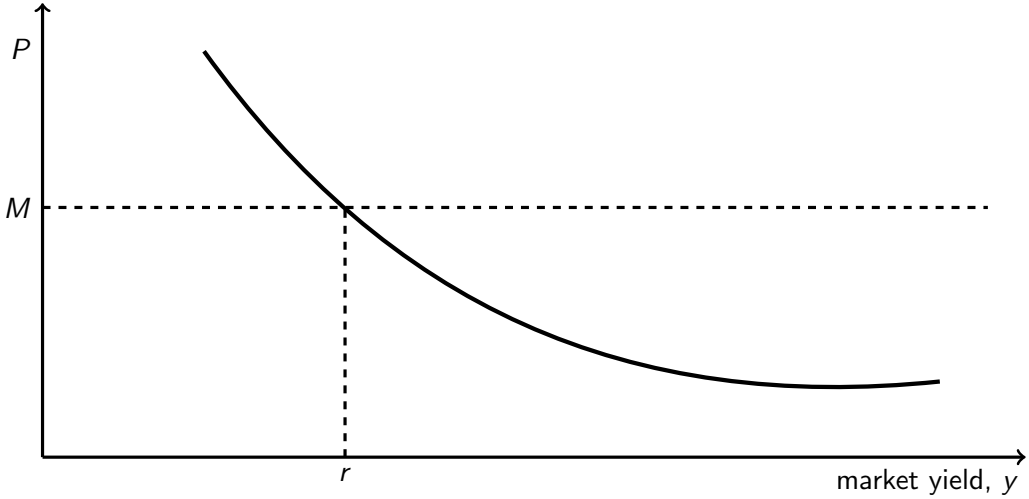
Price-Yield Relationships for Callable and Putable Bonds



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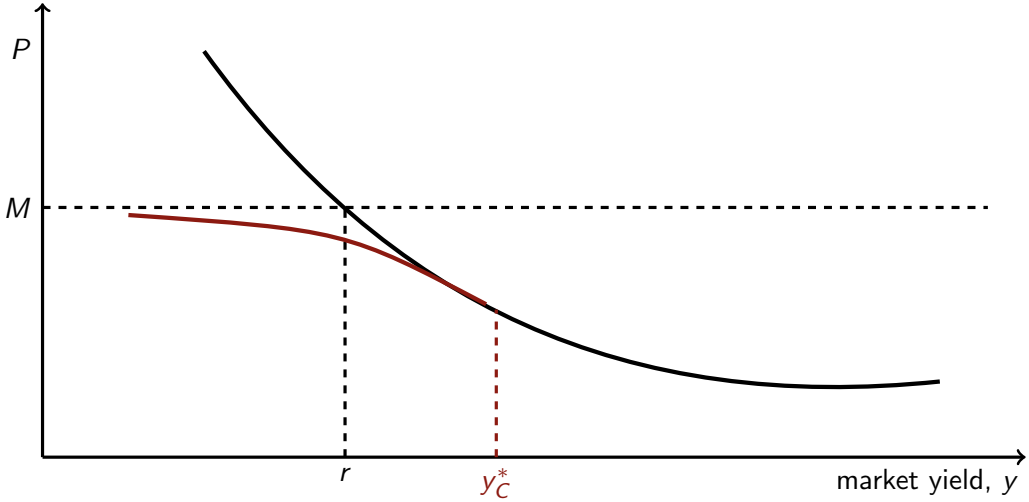

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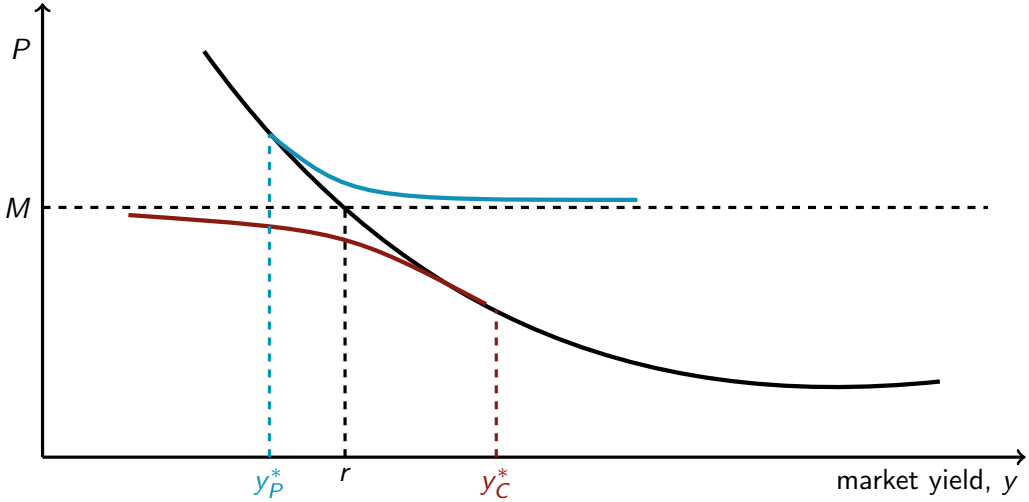

Next time: Calculating Internal Yields

- Today focused on valuing a bond using observable characteristics
  - Annuity + PV of principal
  - Prices and yields are inversely related
  - Nuance such as embedded options or floating rates make pricing more complicated

- **Next time:** What is the actual *return* on a bond, given its price?
  $\rightarrow$ Internal "yield"
  - Matters for interest rate risk — may get to this